Capacity Expansion Network Optimization Report

Model Type: **capacity_expansion**

Metric	Value
Status	Optimal
Objective Value	$49.00
Total Revenue	$70.00
Total Expansion Cost	$11.00
Total Installation Cost	$10.00
Total Cost	$21.00
Net Profit	$49.00
Accepted Demands	2 / 2
Rejected Demands	0 / 2
Acceptance Rate	100.0%
Nodes	3
Edges	2
Total Initial Capacity	3.00
Total Expansion	4.00
Capacity Increase (%)	133.3%

Insights & Highlights

Key Insight	Detail
Most Congested Edge	S -> A (100.0%)
Number of Highly Utilized Edges (≥90%)	3
Most Capacity Expansion	('S', 'T')
Total Lost Revenue	$0.00
Number of Expanded Edges	2

WARNING: Congestion risk in the following edges (utilization ≥90%):
- Edge S -> A: 100.0% used
- Edge A -> T: 100.0% used
- Edge S -> T: 100.0% used

Accepted Demands & Flow Paths

Demand 0: S -> T

Bandwidth: 3.0 units **Revenue:** $40.0
Revenue/BW: $13.33/unit

Path	Fraction	Bandwidth
S -> T	33.33%	1.00
S -> A -> T	66.67%	2.00

Demand 1: S -> T

Bandwidth: 2.0 units **Revenue:** $30.0
Revenue/BW: $15.00/unit

Path	Fraction	Bandwidth
S -> T	100.00%	2.00

Rejected Demands

No demands were rejected.

Capacity Expansions

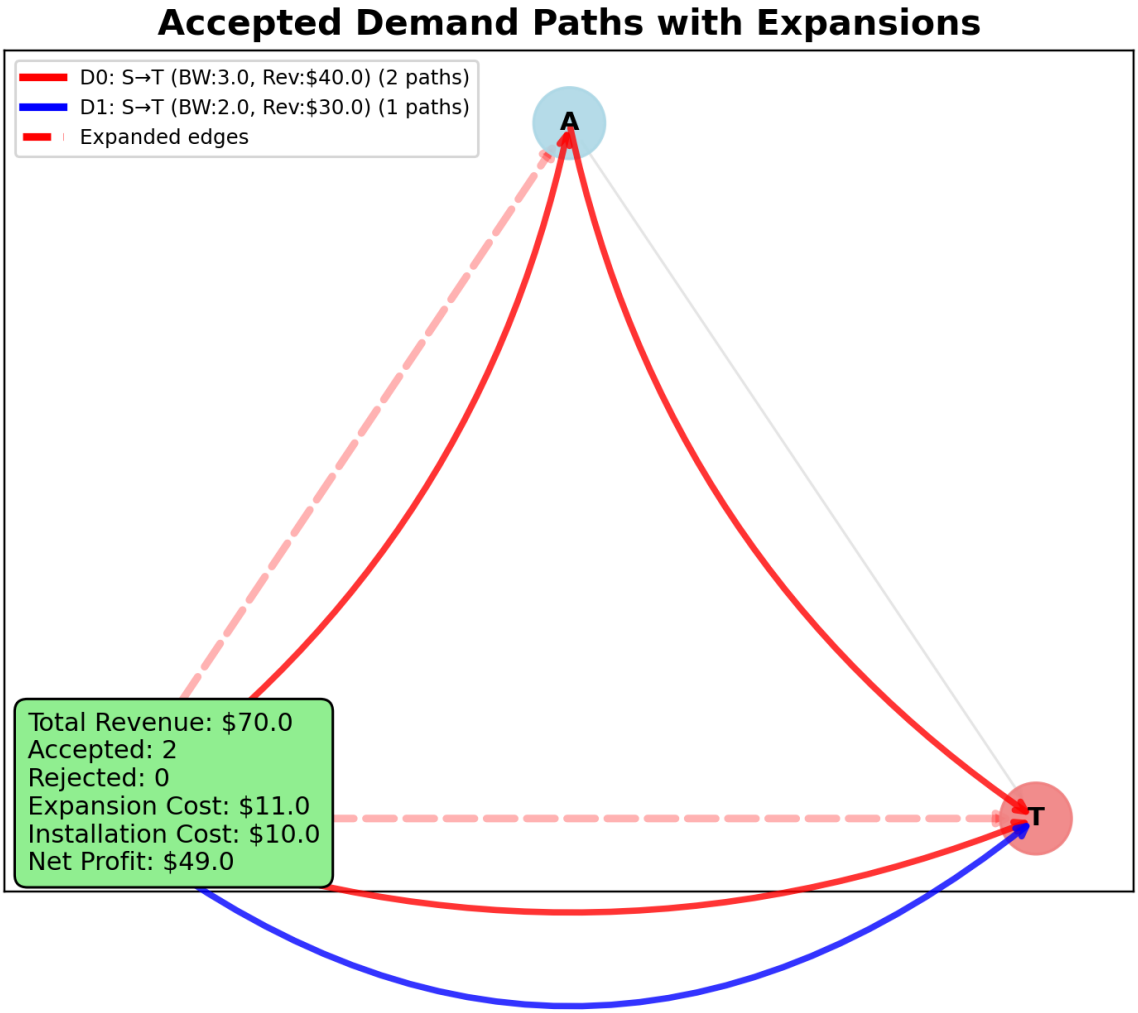
Edge	Expansion (units)	New Capacity	Total Cost ($)
S -> A	1.00	2.00	$2.00
S -> T	3.00	3.00	$19.00

Edge Utilizations (Top 12 Edges)

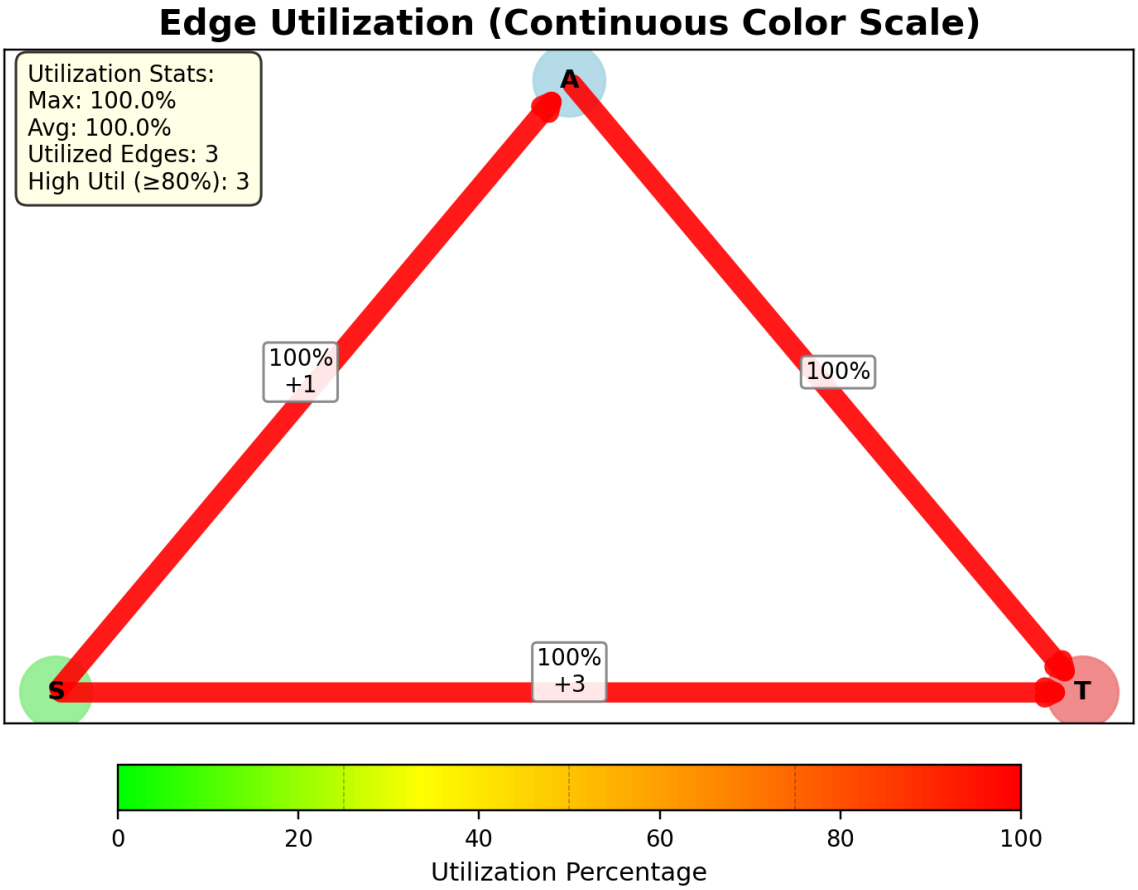
Edge	Flow	Total Capacity	Utilization (%)
S -> A	2.00	2.00	100.0 %
A -> T	2.00	2.00	100.0 %
S -> T	3.00	3.00	100.0 %

Network Analysis Visualizations

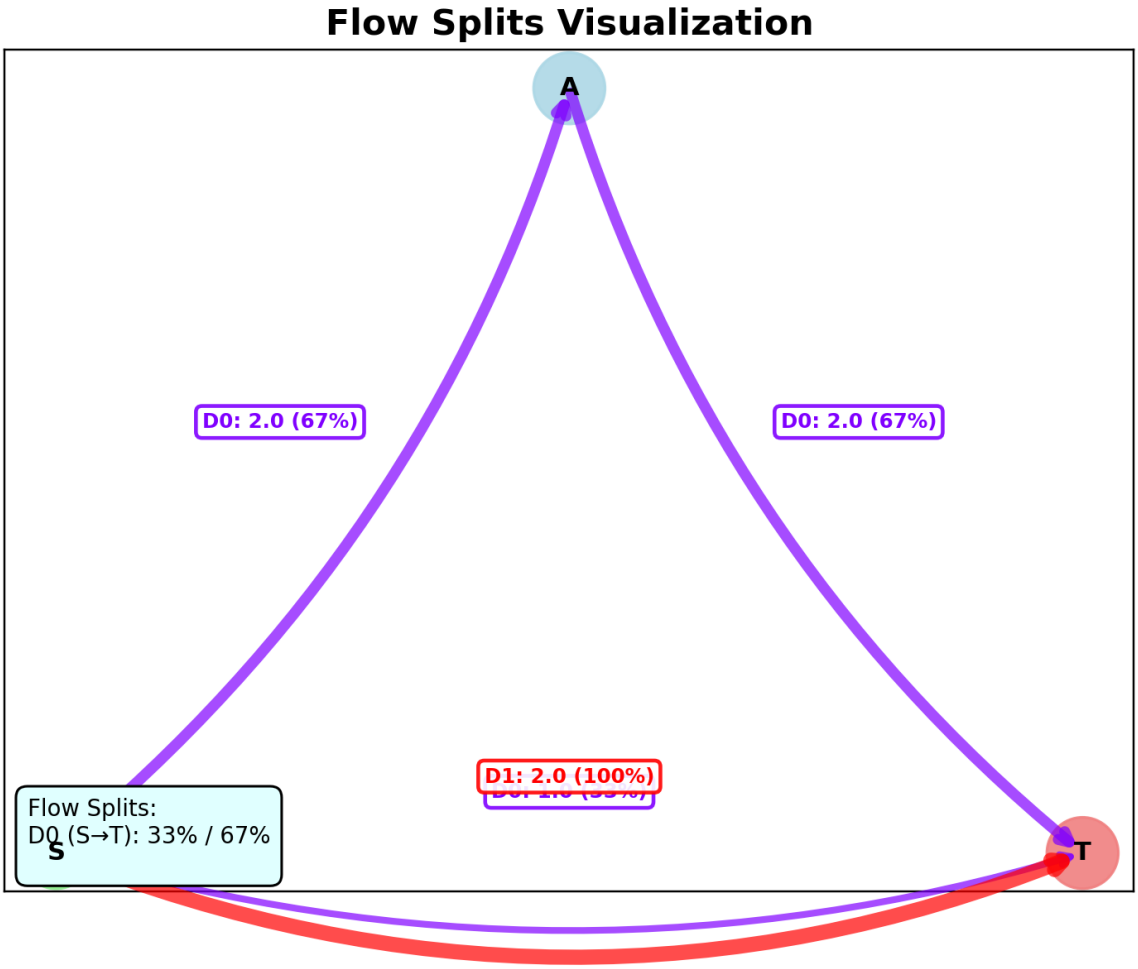
Accepted Demand Paths



Edge Utilization



Flow Splits


Flow Splits Visualization
A
D0: 2.0 (67%)
D0: 2.0 (67%)
D1: 2.0 (100%)
Flow Splits:
D0 (S→T): 33% / 67%
S
T

Capacity Expansion Details

